# Bart Olsthoorn

Ph.D. in Physics · Product Engineer at Flower · Founder of Glimt

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## Profile

Full-stack software engineer with a physics Ph.D., building product end-to-end: data pipelines on Databricks, backend APIs, and the web interfaces on top. Builds much of it with AI agents as a development tool. Equally comfortable leading a small team or training a model.

## Experience

2026 – now **Glimt**
Founder · Sweden
Building scientific instruments, starting with a scanning tunnelling microscope for atomic-resolution imaging.

2023 – now **Flower**
Stockholm · Python, Go, AWS, Databricks

Feb 2026 – now — Product Owner / Product Engineer · Distributed Assets
Owning roadmap and prioritisation for the team; building our systems with AI coding agents.

Dec 2024 – Feb 2026 — Software Engineer · Distributed Assets
Built Flower Bridge, the integration platform for distributed energy assets (residential and C&I batteries, solar, EVs, heat pumps), end-to-end: Databricks data pipelines, backend APIs, and web interfaces.

Jan 2024 – Dec 2024 — Engineering Manager (interim) · Asset Management
Technical recruiting and team leadership during a growth phase.

Feb 2023 – Jan 2024 — Software Engineer · Asset Management
Built a platform supporting the Nordic electrical grid with grid-scale solar and battery systems.

2012 – 2017 **Freelance Software Engineer**
Stockholm, Sweden and Leiden, The Netherlands
Shipped production software across a range of stacks and domains: early generative AI with PyTorch / GANs at *Similar.ai*; Ruby and Elixir work on search, recommendations, and deduplication at *Universal Avenue* (now Velory); scalable Rails SaaS at *InnerBalloons* (acquired by Yext) and web-data systems at *Pointer Brand Protection*.

## Education

2018 – 2023 **Ph.D., Physics · KTH Royal Institute of Technology**
Thesis: *Homology and machine learning for materials informatics.* 11 peer-reviewed publications.

2015 – 2018 **M.Sc., Computational Physics · Stockholm University**

## Selected publications

2023 **Persistent homology of quantum entanglement**
Physical Review B · Editors' Suggestion · link

2019 **Band gap prediction for large organic crystal structures with machine learning**
Advanced Quantum Technologies · link

Full list on Google Scholar.

## Open source

550+ stars across repositories on github.com/bartolsthoorn. Highlights: NVDSP, an iOS / macOS audio DSP library (418★); NQS-numpy, neural-network quantum states in NumPy (35★); simhash-rs, a Rust implementation of SimHash (25★); gohighs, Go bindings for the HiGHS optimizer.

## Skills & languages

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|---|---|
| Core | Python, Go, full-stack web (APIs & interfaces), AWS, Databricks, PostgreSQL, distributed systems, product engineering |
| AI / ML | Agents and agentic loops in production; PyTorch; Pandas / NumPy / scikit-learn; model training & evaluation |
| Past | Ruby / Rails, Elixir, Swift / Objective-C |
| Research | Machine learning for materials, topological data analysis, quantum systems |
| Languages | English, Dutch (native), Swedish (working) |